CS 428

11/9/2026

Luke Bird and Jayden Stewart

Project 4 – Password Manager

For Project 4, we were required to research a selected password manager and replicate its functionality including cryptographically secure random password generation and secure persistent storage. For this project, we are emulating the password manager “LastPass”.

For this project we decided to use a mixture of HTML, CSS, and Javascript. These languages were chosen because they are the most common languages of web applications and were well-suited for our project. Most password managers are primarily used as browser extensions, so this was fitting.

Alongside the ability to securely store passwords, an importaant functionality of LastPass is the ability to generate secure passwords. These passwords can be customized to meet particular length requirements and to include specific characters. The options available in LastPass are as follows:

- Lowercase Letters
- Uppercase Letters
- Numerical Characters
- Special Characters including @^%#*!
- Character Count
    - For free accounts: 1-50 characters
    - For paid accounts: 1-100 characters

Our password manager has all of the same customizations, only including ‘$’ and ‘&’ as special character possibilities in addition to those avaliable on LastPass. Our password generator allows a selection of characters ranging from 1-100.

Our password generator also includes the ability to rate the strength of the inputted and generated passwords. Passwords are rated on a scale of Very Weak, Weak, Moderate, Strong, and Very Strong. The strength of the password is determined by a point system, with passwords gaining points from extra length and more diverse characters and losing points for including common password phrases. Suggestions to improve a password are offered according to its detected deficiencies.

The randomness in our password generator comes from the Cryto web API included in most modern browsers. This widely-used library is useful to generate secure random numbers. In our password manager we use this library to repeatedly generate random numbers with the ‘getRandomValues()’ function. These numbers are used to select a character out of the list of possible characters selected by the user.

Our password manager’s interface is simple and straightforward. Upon opening the page, the user is prompted to choose a Master Password of at least 12 characters. This ensures that the master password remains secure while providing convenient access. At this point the user also has the option to instead import a backup of the password manager’s database. This will restore the password manager to the point at which the backup was created. After the selection of a master password, the user will be prompted to sign in using that

password In the future rather than creating a new one. We chose not to include the ability to ‘reset’ a forgotten master password in order to eliminate a potential security vulnerability.

At the bottom of the page is the password generator interface. We decided that the password generator should be available even on the master password creation page to assist users in creating a more secure master password.

Once a user has selected a master password or signed in, the gain access to the main Password Vault page. This is where passwords can be saved for later use. The page has three input boxes: one for the website the login information corresponds to, one for the username field, and one for the password. We decided that a cooresponding website should not be required in order to save a password because this offers more possible use cases.

The user is also presented with four buttons. The save button will save the password and username combination to the vault. The lock button sign the user out and requires the master password to be input again before moving on. The “Export Encrypted Backup” option generates a securely encrypted backup file of the entire password database. The Import Backup button takes one of thes beackup files and overwrites the current database with the data from the backup.

Finally, the user is presented a list of all username and password combinations that have been saved in the vault. The user also sees the Password Generator below this.

Our password manager performs very similarly to the LastPass tool. We include all of the same customization options and a similar ability to store

passwords securely for later use. We also include several quality of life features that are not present in the LastPass password manager like password strength evaluation, several more special character options, and the ability to use the password generator to create your master password.

In conclusion, our password manager was a success. It hold up when compared to the LastPass password manager and uses good security both for randomness and encryption.

AI Usage Statement: Artificial Intelligence was used exclusively for brainstorming purposes. Artificial Intelligence was not used or consulted in any capacity during any other phase of the making of this project.